Numerical Workflow for Surface-Electrode Ion Trap Design

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Overview

This work presents a numerical workflow for the design and optimization of surface-electrode ion traps. The pipeline integrates electrostatic modeling, parameter sweeps, and physical scaling to produce hardware-relevant trap characteristics.

geometry → Laplace solve → electric field → pseudopotential → trap detection → optimization → physical metrics

Outputs include trap height (μm), pseudopotential scale (eV), and secular frequencies (MHz).

Method

Electrostatics

We solve:

$$\nabla^2 V = 0$$

Field

$$\mathbf{E} = -\nabla V$$

Pseudopotential

$$\Phi_{\text{pseudo}} = \frac{q^2 |\mathbf{E}|^2}{4m\Omega^2}$$

Trap Condition

$$\lambda_i > 0$$

Objective

$$\text{score} = \frac{\text{trap strength}}{\text{trap height}^2}$$

Results

The optimization produces a physically consistent trapping configuration. The pseudopotential map (Fig. 1) shows a stable minimum above the electrode surface.

- Trap height: $\sim 30\,\mu$m
- Secular frequency: $\sim$ 1–2 MHz
- Stable quadratic confinement near the trap point

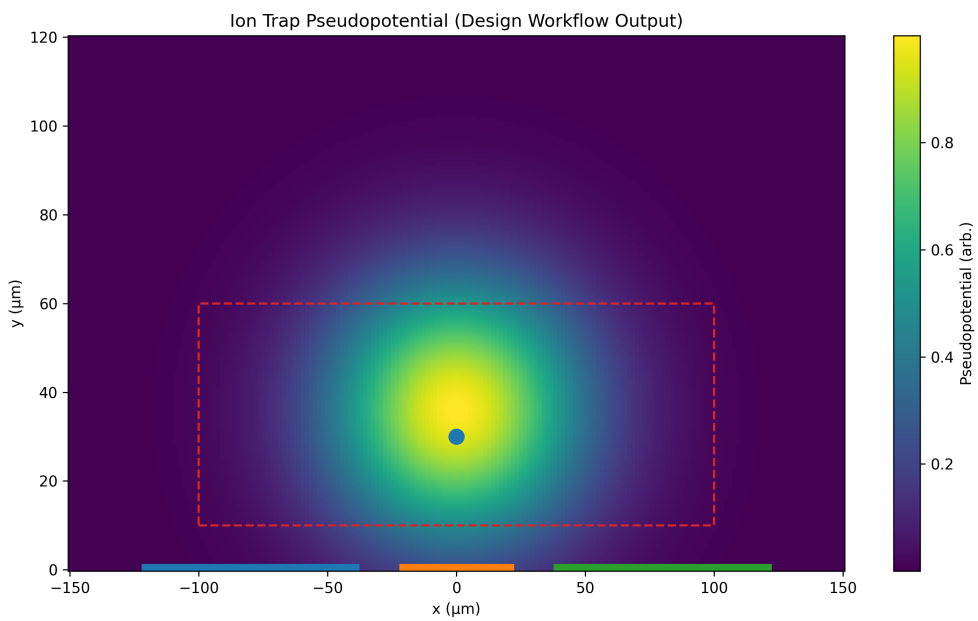

Figure 1: RF pseudopotential map showing electrode geometry (bottom), optimized trap location (blue point), and search window (dashed). The trap forms as a stable minimum above the RF electrode.

Interpretation

The results reflect expected surface-trap behavior:

- Smaller electrode spacing increases electric field gradients and confinement strength
- Larger geometries raise the trap height but reduce curvature
- Optimal designs balance confinement strength against practical trapping distance

The objective function

$$\text{score} = \frac{\text{trap strength}}{\text{trap height}^2}$$

encodes this tradeoff, favoring strong confinement while penalizing traps that form too far from the surface.

Conclusion

This workflow provides:

- Electrostatic modeling
- Multi-parameter optimization
- Curvature-based trap validation
- Conversion to physical units

It mirrors early-stage trapped-ion design and produces realistic trap metrics.

Code: `https://github.com/thinkthoughts/ion-trap-parameter-lab`